MODULE 19 · DOMAIN 4 - EVALUATION, TESTING AND OPTIMISATION

A/B Testing and Iteration

When to test online, how to design an experiment that can be trusted, the pitfalls that fake a result, and deciding to promote, reject or iterate.

DOMAIN 4 of 7 EXAM WEIGHT 16% MODULE 19 of 38

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An A/B test is a clinical trial for a change. **Decide what would convince you before you start.**

Offline evaluation or online experiment?

OFFLINE EVALUATION

Is it safe to try?

- A fixed set, before release
- Fast, cheap, repeatable
- Catches regressions
- No user is exposed
- Cannot show real-world outcomes

ONLINE A/B TEST

Does it actually help?

- Real traffic, real outcomes
- Needs enough traffic and time
- Exposes some users to the change
- Needs guardrail metrics
- Shows behaviour no test set predicted

THE TIEBREAKER

Offline first, always. Only changes that pass offline earn an online test.

Designing an experiment, in order

01
Hypothesis
One change, one expected effect.

02
Metrics
One primary, plus guardrails.

03
Power
Smallest effect worth acting on; sample size.

04
Randomise, run
Random assignment, fixed plan.

05
Decide
Promote, reject or iterate.

06
Document
Hypothesis, result, decision.

ON THE EXAM

Everything before randomise is decided in advance. The exam asks which steps come before the run.

The parts of a sound experiment

PART	WHAT GOOD LOOKS LIKE
Hypothesis	Caching cuts p95 first-token time by a third, with no accuracy loss
Primary metric	One, chosen in advance, tied to the requirement
Guardrail metrics	Accuracy, safety, cost, escalations: must not worsen
Minimum detectable effect	The smallest difference worth acting on; sets the sample size
Random assignment	By user or conversation, never by shift, team or week
Stopping rule	A fixed sample or duration, set before the start

ON THE EXAM

Decide before you start how small a difference you would care about. That decides how long you must run.

Pitfalls that fake a result

- **Peeking:** stopping as soon as the difference first looks significant.
- **Two changes at once:** you cannot say which one caused the effect.
- **Non-random groups:** shifts, teams or weeks differ in ways you did not measure.
- **Fishing:** checking many metrics and reporting whichever moved.
- **Too little traffic:** a test too small to detect the effect, read as no effect.

> ON THE EXAM
>
> Every pitfall here produces a confident answer that is wrong. The design protects you from yourself.

What is different about testing LLM systems

- **Outputs vary:** the same input can give different answers, so plan for more samples.
- **Quality needs grading:** score sampled live outputs with calibrated graders, per arm.
- **Pin every version** in both arms: model, prompt, tools, index. Nothing drifts mid-test.
- **Track cost and latency per arm,** not just quality.
- **Shadow first** when exposure is risky: run the new version silently, compare offline.

> ON THE EXAM
>
> An LLM experiment needs the eval machinery running live: graders, versions and per-arm costs.

Deciding what to do with the result

RESULT	DECISION
Primary improves, guardrails hold	Promote, with a staged rollout
Primary flat or worse	Reject, and record why
Mixed, or too small to tell	Iterate: refine offline, or extend if planned
A guardrail is breached	Stop and roll back, whatever the primary says

ON THE EXAM

A guardrail breach outranks a primary win. Faster and wrong is not an improvement.

Iteration that compounds

- **Live failures become eval cases,** so the offline set keeps learning.
- **One change per iteration,** each with its own hypothesis.
- **Staged exposure:** shadow runs, then a small share of traffic, then wider.
- **A decision log:** what was tried, what happened, what was decided.
- **Re-run the regression suite** before every promotion.

> ON THE EXAM
>
> Each loop should leave a better eval set and a clearer record, not just a new version.

> Tallyhawk has a new support prompt intended to raise first-contact resolution. The product lead wants it live for half of customers next week.
>
> Corvane is asked what must happen before the experiment starts.

What should come first?

A Pass the offline regression suite, then write the hypothesis, primary and guardrail metrics, and sample size

B Launch to half of customers and see which metrics improve

C Choose the primary metric after the first week's results arrive

D Run it for a single day to get a quick answer

◆ PAUSE AND DECIDE

What should come first?

A Pass the offline regression suite, then write the hypothesis, primary and guardrail metrics, and sample size

Safe to try, then decide in advance what would convince you.

B Launch to half of customers and see which metrics improve

Untested exposure, and fishing for a result.

C Choose the primary metric after the first week's results arrive

A metric chosen after the fact proves nothing.

D Run it for a single day to get a quick answer

Duration must come from the sample size, not impatience.

THE SIGNAL

Before an A/B run: offline pass, hypothesis, metrics and power. All decided in advance.

> Mereside wants to test a new agent-assist prompt in its Preston contact centre.
>
> The operations manager proposes the morning shift uses the new version and the afternoon shift the old one, since the two shifts are the same size.

What should Corvane recommend?

A Accept the plan, since the two groups are equal in size

B Give the new version to the most experienced team, for better feedback

C Assign advisers to versions at random, so shift, experience and call mix are balanced

D Run the new version for one week, and the old version the week after

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Accept the plan, since the two groups are equal in size

Equal size, different calls. Mornings and afternoons differ.

B Give the new version to the most experienced team, for better feedback

Experience, not the prompt, may drive the result.

C Assign advisers to versions at random, so shift, experience and call mix are balanced

Randomisation balances what you did not think to measure.

D Run the new version for one week, and the old version the week after

Weeks differ: campaigns, rate changes, season.

THE SIGNAL

Groups chosen by shift, team or week are confounded. Randomise.

> Quillon expects a portal change to raise first-contact resolution by about one percentage point. The portal handles about 4,000 conversations a week.
>
> A power calculation shows two weeks of traffic could only detect a change of five points or more.

What should Corvane recommend?

A Set the duration from the power calculation, or rely on offline evidence if that is not feasible

B Run for two weeks and treat any difference as real

C Stop the test as soon as the difference first looks significant

D Run for two weeks, and read no difference as proof of no effect

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Set the duration from the power calculation, or rely on offline evidence if that is not feasible

Run long enough to see the effect you care about, or do not pretend to.

B Run for two weeks and treat any difference as real

Small differences at this size are mostly noise.

C Stop the test as soon as the difference first looks significant

Peeking inflates false wins.

D Run for two weeks, and read no difference as proof of no effect

Too small to detect it is not the same as no effect.

THE SIGNAL

The smallest effect worth acting on sets the sample size. An underpowered test answers nothing.

> In Tallyhawk's A/B test, the new support agent cut handling time by a fifth, the primary metric.
>
> In the same test, escalations to human staff rose, and refund errors doubled. Both were guardrail metrics.

What should Corvane recommend?

A Promote it, because the primary metric improved

B Promote it to half of customers as a compromise

C Do not promote; the guardrail breach overrides the primary win, so investigate

D Extend the test to collect more data on handling time

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Promote it, because the primary metric improved
Ignores the guardrails, which exist for exactly this.

B. Promote it to half of customers as a compromise
Doubles refund errors for half the customers.

C. Do not promote; the guardrail breach overrides the primary win, so investigate
Faster and wrong is not better.

D. Extend the test to collect more data on handling time
The question was answered; the harm is real now.

THE SIGNAL

A guardrail breach outranks a primary win. Stop, investigate, iterate offline.

> Corvane wants to know whether a newer model version writes better discharge summaries for Quillon. Clinicians review and sign every summary.
>
> A developer suggests switching half of live patients to the new version straight away, to get real data fast.

What should Corvane recommend?

A. Switch half of live patients now, since clinicians review everything anyway

B. Run the evaluation set offline first; only if it passes, run a controlled live trial with clinician review continuing

C. Switch everyone, since newer models are generally better

D. Ask two clinicians which of two sample summaries they prefer

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Switch half of live patients now, since clinicians review everything anyway
Review is a safety net, not a test plan.

B. Run the evaluation set offline first; only if it passes, run a controlled live trial with clinician review continuing
Offline proves it safe to try; the trial proves it helps.

C. Switch everyone, since newer models are generally better
Untested, and newer is not always better for your task.

D. Ask two clinicians which of two sample summaries they prefer
Anecdote, not evidence.

THE SIGNAL

Offline first, always. A human safety net does not replace testing.

> Tallyhawk's experiment report says: satisfaction, the pre-registered primary metric, rose significantly on day three, so the test was stopped.
>
> It adds that the retrieval index was also rebuilt halfway through. Users were assigned at random.

Which two flaws most undermine the result? Choose two.

A Users were assigned to versions at random

B The test stopped as soon as the result first looked significant

C The retrieval index changed during the test

D The primary metric was chosen before the test began

E The test included customers in both the UK and the US

◆ PAUSE AND DECIDE

Which two flaws most undermine the result? Choose two.

A Users were assigned to versions at random

Correct practice, not a flaw.

B The test stopped as soon as the result first looked significant

Peeking: early wins are often noise.

C The retrieval index changed during the test

A second change, mid-test: the effect cannot be attributed.

D The primary metric was chosen before the test began

Correct practice, not a flaw.

E The test included customers in both the UK and the US

Fine with random assignment; it widens coverage.

THE SIGNAL

Stopping early and changing something mid-test are the classic ways to fake a result.

In five lines

- **Offline first:** only changes that pass the eval set earn a live test.
- **Design in advance:** hypothesis, one primary, guardrails, effect size, sample size.
- **Randomise** by user or conversation, never by shift, team or week.
- **Avoid the fakes:** peeking, two changes, fishing, too little traffic.
- **Guardrail breaches outrank primary wins.** Promote, reject or iterate, and document.

COMING UP

Next: diagnosing failures. Prompt, retrieval, hallucination or model mismatch?